

Performance Report

Website Performance Report

Course: Web Programming

Project: EventsX Website

Comparison Website: Youm7 (youm7.com)

1. Introduction

Website performance is a critical factor that affects user experience, usability, and search engine optimization (SEO). Fast-loading websites improve user satisfaction, reduce bounce rates, and provide smoother interactions.

This report analyzes and compares the performance of:

- EventsX – a locally developed website created as part of the Web Programming course.
- Youm7 – a large-scale, real-world news website with heavy content and third-party integrations.

The evaluation is based on Chrome DevTools Performance Insights, focusing on key performance metrics such as LCP, INP, CLS, and overall resource usage. Screenshots from the performance analysis are included in the appendix.

2. Performance Metrics Used

The following metrics were used to evaluate both websites:

2.1 Largest Contentful Paint (LCP)

- Measures how long it takes for the main content to load.
- Ideal value: < 2.5 seconds

2.2 Interaction to Next Paint (INP)

- Measures responsiveness to user interactions.
- Ideal value: < 200 ms

2.3 Cumulative Layout Shift (CLS)

- Measures visual stability of the page.
- Ideal value: < 0.1

2.4 Resource and Rendering Analysis

- JavaScript execution time
 - Rendering and painting cost
 - Network requests
 - Third-party scripts
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3. Performance Analysis of Youm7 Website

Based on the Chrome DevTools screenshots:

3.1 Observed Metrics

- LCP: ~0.42 seconds
- INP: ~9 ms
- CLS: ~0.05

3.2 Performance Characteristics

- The website loads a large number of images, ads, and external scripts.
- Heavy use of third-party services such as ads, analytics, and tracking tools.
- Complex layout with continuous dynamic content updates.
- Higher rendering and scripting costs due to advertisements and media elements.

3.3 Strengths

- Optimized images and font loading.
- Acceptable LCP despite large content size.
- Stable layout with controlled layout shifts.

3.4 Weaknesses

- High dependency on third-party scripts.
- Increased network requests.
- Higher CPU usage during rendering.

(Screenshot 1: Youm7 Performance Insights)

4. Performance Analysis of EventsX Website

EventsX was tested locally using 127.0.0.1, ensuring no network latency.

4.1 Observed Metrics

- LCP: Between 0.08 – 0.15 seconds
- INP: Between 9 – 12 ms
- CLS: 0

4.2 Performance Characteristics

- Minimal and well-structured HTML and CSS.
- Limited JavaScript usage.
- No third-party ads or tracking scripts.
- Optimized layout with stable content loading.

4.3 Strengths

- Extremely fast load time.
- Zero layout shift, providing a stable user experience.

- Low scripting and rendering overhead.
- Clean and maintainable code structure.

4.4 Weaknesses

- Limited real-world stress testing (local environment).
- No CDN or production-level optimization tested yet.

(*Screenshot 2 and 3: EventsX Performance Insights*)

5. Comparative Analysis

Metric	Youm7	EventsX
LCP	~0.42 s	~0.08–0.15 s
INP	~9 ms	~9–12 ms
CLS	0.05	0
Third-party Scripts	High	None
Page Complexity	Very High	Low
Ads & Media	Extensive	Minimal

Discussion:

- EventsX outperforms Youm7 in raw performance metrics due to its simplicity and lack of heavy content.
 - Youm7, despite its complexity, maintains acceptable performance through advanced optimizations.
 - The comparison highlights the impact of third-party scripts and media-heavy layouts on performance.
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6. Conclusion

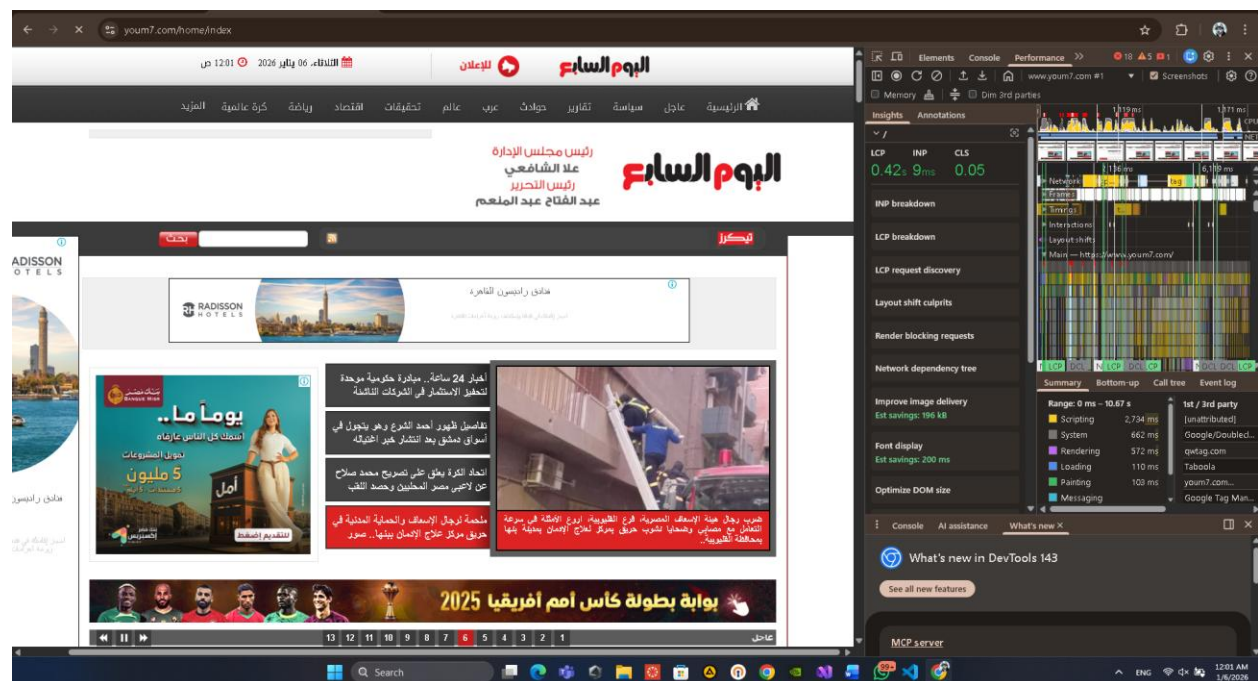
The performance analysis demonstrates that EventsX achieves excellent performance results, especially in terms of loading speed, responsiveness, and layout stability. This is mainly due to:

- Clean code structure
- Minimal JavaScript usage
- Absence of third-party scripts

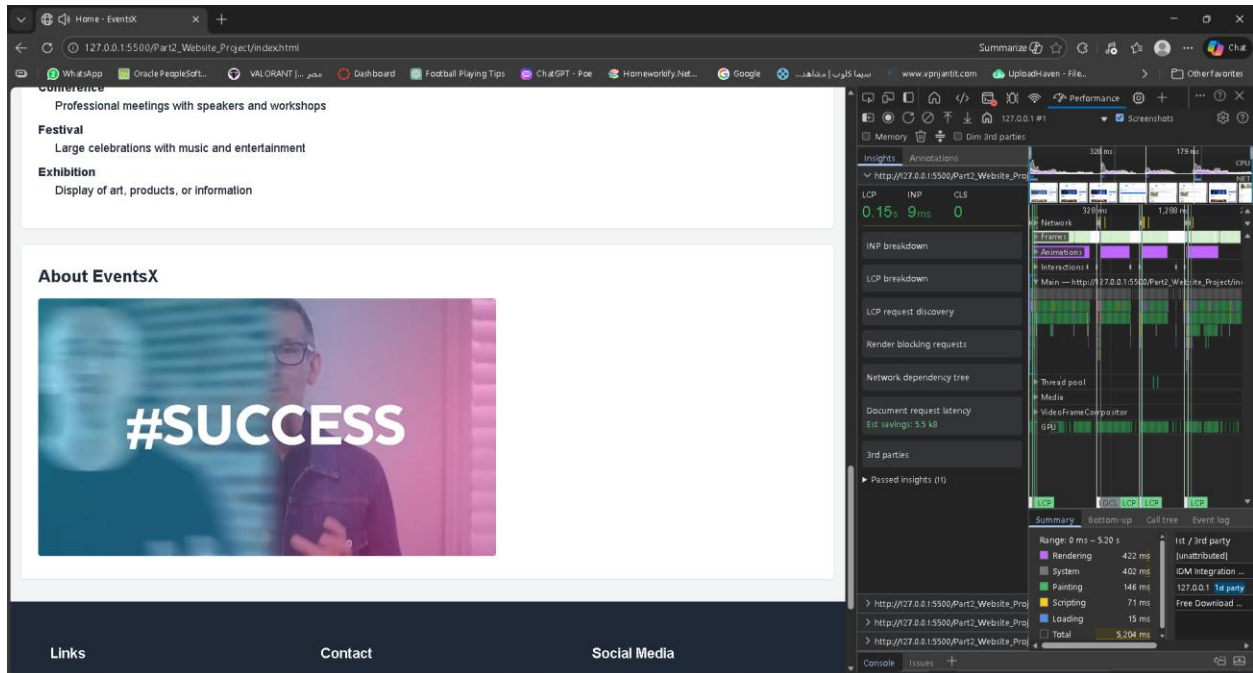
On the other hand, Youm7, as a real-world commercial website, handles significantly more content and complexity, which naturally impacts performance.

Overall, this project shows a strong understanding of web performance optimization principles and effective use of Chrome DevTools for performance analysis.

Screenshot 1



Screenshot 2



Screenshot 3

